

P1: Graphs & Grids

- ☐ 1.1 Multi-Source BFS (Grids / Unweighted Graphs) ☐ **MUST DO (Warsaw L4)**
- ☐ 1.2 0-1 BFS (Shortest Path with Edge Weights 0 or 1)
- ☐ 1.3 Dijkstra's Algorithm (Non-Negative Weighted Shortest Path)
- ☐ 1.4 Topological Sort & Cycle Detection
- ☐ 1.5 Disjoint Set Union (DSU / Union-Find)
- ☐ 1.6 Bipartite Graph Verification (2-Coloring)

P2: Trees, LCA & Trie

- ☐ 2.1 Lowest Common Ancestor (LCA)
- ☐ 2.2 Tree DP: Path Sum & Diameter Pattern
- ☐ 2.3 Binary Tree Serialization & Deserialization
- ☐ 2.4 Prefix Tree (Trie)

P3: Binary Search & Feasibility Space

- ☐ 3.1 The Universal Binary Search Templates
- ☐ 3.2 Lower Bound & Upper Bound (Exact Indexing)
- ☐ 3.3 Search in Rotated Sorted Array

P4: Dynamic Programming & State Machines

- ☐ 4.1 Knapsack Patterns
- ☐ 4.2 Longest Increasing Subsequence (LIS) in $\mathcal{O}(N \log N)$

P5: Monotonic Stack & Monotonic Deque

- ☐ 5.1 Next Greater Element (Template)
- ☐ 5.2 Largest Rectangle in Histogram ($\mathcal{O}(N)$)

P6: Prefix Sum, Two Pointers & Sliding Window

- ☐ 6.1 Dynamic Sliding Window (Universal Structure)
- ☐ 6.2 The "Exact $\mathcal{K}$ " Subarray Trick
- ☐ 6.3 3-Sum (Duplicate Skipping Pointers)
- ☐ 6.4 1D Prefix Sum + HashMap Complement (Subarray Sum Equals K)
- ☐ 6.5 2D Prefix Sum Formula (Matrix Range Query in $\mathcal{O}(1)$)
- ☐ 6.6 Difference Array (1D Range Updates in $\mathcal{O}(1)$) ☐ **MUST DO (Warsaw L4)**

P7: Intervals & Sweep Line

- ☐ 7.1 Merge Intervals ☐ **MUST DO (Warsaw L4)**
- ☐ 7.2 Meeting Rooms II / Sweep Line Algorithm ☐ **MUST DO (Warsaw L4)**

P8: Heaps, Top-K & QuickSelect

- ☐ 8.2 QuickSelect ($O(N)$ Average, $O(1)$ Extra Space)

P9: Backtracking & State-Space Pruning

- ☐ 9.1 Combinations / Subsets with Duplicate Pruning
- ☐ 9.2 Permutations with Duplicates
- ☐ 9.3 In-Place Grid Backtracking (Word Search)

P10: Bit Manipulation & Core Math

P11: Google L4 Edge Cases & Java Speed Cheat-Sheet

- ☐ 11.1 Fast Java Standard Library Constructs
- ☐ 11.2 Google L4 Verification & Edge Case Checklist
- ☐ 11.3 Google Warsaw L4 Hub Specifics & Core Signal Guide

Part 2: Question Bank (neetcode150.md)

1. Arrays and Hashing

- ☐ Q1: Two Sum (LC 1)
- ☐ Q2: Group Anagrams (LC 49)
- ☐ Q3: Top K Frequent Elements (LC 347)
- ☐ Q4: Product of Array Except Self (LC 238)
- ☐ Q5: Valid Sudoku (LC 36)
- ☐ Q6: Longest Consecutive Sequence (LC 128)
- ☐ Q7: Encode and Decode Strings (LC 271)

2. Two Pointers

- ☐ Q1: Two Sum II - Input Array Is Sorted (LC 167)
- ☐ Q2: 3Sum (LC 15)
- ☐ Q3: Container With Most Water (LC 11)
- ☐ Q4: Trapping Rain Water (LC 42)

3. Prefix Sum & Difference Arrays (Google Core)

- ☐ Q1: Subarray Sum Equals K (LC 560)
- ☐ Q2: Contiguous Array (LC 525)
- ☐ Q3: Continuous Subarray Sum (LC 523)
- ☐ Q4: Subarray Sums Divisible by K (LC 974)
- ☐ Q5: Range Sum Query 2D - Immutable (LC 304)
- ☐ Q6: Corporate Flight Bookings / Range Addition (LC 1109 / LC 370)
- ☐ Q7: Shortest Subarray with Sum at Least K (LC 862) ☐ **MUST DO (Warsaw L4)**

4. Sliding Window

- ☐ Q1: Best Time to Buy and Sell Stock (LC 121)
- ☐ Q2: Longest Substring Without Repeating Characters (LC 3)
- ☐ Q3: Longest Repeating Character Replacement (LC 424)
- ☐ Q4: Permutation in String (LC 567)
- ☐ Q5: Minimum Window Substring (LC 76) ☐ **MUST DO (Warsaw L4)**

5. Stack

- ☐ Q1: Min Stack (LC 155)
- ☐ Q2: Evaluate Reverse Polish Notation (LC 150)
- ☐ Q3: Generate Parentheses (LC 22)
- ☐ Q4: Daily Temperatures (LC 739)
- ☐ Q5: Car Fleet (LC 853)
- ☐ Q6: Largest Rectangle in Histogram (LC 84)

6. Binary Search

- ☐ Q1: Search a 2D Matrix (LC 74)
- ☐ Q3: Find Minimum in Rotated Sorted Array (LC 153)
- ☐ Q4: Search in Rotated Sorted Array (LC 33)
- ☐ Q5: Time Based Key-Value Store (LC 981)

7. Linked List

- ☐ Q1: Reorder List (LC 143)
- ☐ Q2: Remove Nth Node From End of List (LC 19)
- ☐ Q3: Copy List with Random Pointer (LC 138)
- ☐ Q4: Add Two Numbers (LC 2)
- ☐ Q5: Find the Duplicate Number (LC 287)
- ☐ Q6: LRU Cache (LC 146) ☐ **MUST DO (Warsaw L4)** ☐ **TRENDING IN POOL**
- ☐ Q7: Merge K Sorted Lists (LC 23)

8. Trees

- ☐ Q1: Diameter of Binary Tree (LC 543)
- ☐ Q2: Lowest Common Ancestor of a BST (LC 235)
- ☐ Q3: Binary Tree Level Order Traversal (LC 102)
- ☐ Q4: Binary Tree Right Side View (LC 199)
- ☐ Q5: Count Good Nodes in Binary Tree (LC 1448)
- ☐ Q6: Validate Binary Search Tree (LC 98)
- ☐ Q7: Kth Smallest Element in a BST (LC 230)
- ☐ Q8: Construct Binary Tree from Preorder and Inorder Traversal (LC 105)
- ☐ Q9: Binary Tree Maximum Path Sum (LC 124)
- ☐ Q11: Step-by-Step Directions From a Binary Tree Node to Another (LC 2096) ☐ **MUST DO (Warsaw L4)** ☐ **TRENDING IN POOL**

TRENDING IN POOL

9. Tries

- ☐ Q1: Implement Trie (Prefix Tree) (LC 208)
- ☐ Q2: Design Add and Search Words Data Structure (LC 211)
- ☐ Q3: Word Search II (LC 212)

10. Graphs

- ☐ Q1: Number of Islands (LC 200)
- ☐ Q2: Max Area of Island (LC 695)
- ☐ Q3: Clone Graph (LC 133)
- ☐ Q4: Rotting Oranges (LC 994)
- ☐ Q5: Pacific Atlantic Water Flow (LC 417)
- ☐ Q6: Surrounded Regions (LC 130)
- ☐ Q7: Course Schedule (LC 207)
- ☐ Q8: Course Schedule II (LC 210)
- ☐ Q9: Redundant Connection (LC 684)
- ☐ Q10: Number of Connected Components (LC 323)
- ☐ Q4: Graph Valid Tree (LC 261)

12. Dynamic Programming (1-D)

- ☐ Q1: House Robber (LC 198)
- ☐ Q2: House Robber II (LC 213)
- ☐ Q3: Longest Palindromic Substring (LC 5)
- ☐ Q4: Palindromic Substrings (LC 647)
- ☐ Q5: Decode Ways (LC 91)
- ☐ Q6: Coin Change (LC 322)
- ☐ Q7: Maximum Product Subarray (LC 152)
- ☐ Q8: Word Break (LC 139)
- ☐ Q9: Longest Increasing Subsequence (LC 300)

- ☐ Q10: Partition Equal Subset Sum (LC 416)
- ☐ Q1: Unique Paths (LC 62)
- ☐ Q2: Longest Common Subsequence (LC 1143)
- ☐ Q3: Best Time to Buy and Sell Stock with Cooldown (LC 309)
- ☐ Q4: Coin Change II (LC 518)
- ☐ Q5: Target Sum (LC 494)

14. Heaps / Priority Queue

- ☐ Q1: Kth Largest Element in a Stream (LC 703)
- ☐ Q2: K Closest Points to Origin (LC 973)
- ☐ Q3: Kth Largest Element in an Array (LC 215)
- ☐ Q4: Task Scheduler (LC 621)
- ☐ Q5: Design Twitter (LC 355)
- ☐ Q7: Snapshot Array (LC 1146) ☐ **MUST DO (Warsaw L4)** ☐ **TRENDING IN POOL**

15. Greedy

- ☐ Q1: Jump Game (LC 55)
- ☐ Q2: Jump Game II (LC 45)
- ☐ Q3: Gas Station (LC 134)
- ☐ Q4: Hand of Straights (LC 846)
- ☐ Q5: Merge Triplets to Form Target (LC 1899)
- ☐ Q6: Partition Labels (LC 763)
- ☐ Q7: Valid Parenthesis String (LC 678)

16. Backtracking

- ☐ Q1: Subsets (LC 78)
- ☐ Q2: Combination Sum (LC 39)
- ☐ Q3: Permutations (LC 46)
- ☐ Q4: Subsets II (LC 90)
- ☐ Q5: Combination Sum II (LC 40)
- ☐ Q6: Word Search (LC 79)
- ☐ Q7: Palindrome Partitioning (LC 131)
- ☐ Q8: Letter Combinations of a Phone Number (LC 17)
- ☐ Q11: Restore IP Addresses (LC 93)
- ☐ Q12: Combination Sum III (LC 216)

17. Intervals

- ☐ Q1: Insert Interval (LC 57)

- ☐ Q2: Merge Intervals (LC 56) ☐ **MUST DO (Warsaw L4)**
- ☐ Q3: Non-overlapping Intervals (LC 435)
- ☐ Q4: Meeting Rooms II (LC 253) ☐ **MUST DO (Warsaw L4)**
- ☐ Q6: My Calendar I & II (LC 729 & LC 731) ☐ **MUST DO (Warsaw L4)** ☐ **TRENDING IN POOL**
- ☐ Q7: Employee Free Time (LC 759) ☐ **MUST DO (Warsaw L4)** ☐ **TRENDING IN POOL**

18. Math & Geometry

- ☐ Q1: Rotate Image (LC 48)
- ☐ Q2: Spiral Matrix (LC 54)
- ☐ Q3: Set Matrix Zeroes (LC 73)
- ☐ Q4: Pow(x, n) (LC 50)
- ☐ Q5: Multiply Strings (LC 43)
- ☐ Q6: Detect Squares (LC 2013)

19. Bit Manipulation

- ☐ Q1: Counting Bits (LC 338)

